How AutoVet Measures Forecast Accuracy

A plain-language guide to R^2, RMSE and MAE, with a worked example

1. The big idea

AutoVet predicts two things for the clinic:

- **Appointments**: how many appointments to expect in the coming weeks.
- **Inventory**: how fast each item is used, and when it will run out.

To make a prediction, AutoVet draws the best straight **trend line** through past data (this is **linear regression**) and extends it into the future.

A prediction is only useful if we know **how much to trust it**. That is what R^2, RMSE and MAE are for. They are the system's "report card".

2. How AutoVet tests itself: the 80/20 check

Think of a student reviewing for an exam. If the exam uses the exact questions they practised, a high score proves nothing. So AutoVet keeps some data **hidden**:

Step 1. Split the history in time order: the oldest **80%** is for learning, the most recent **20%** is hidden.
Step 2. Draw the trend line using only the 80%.
Step 3. Use that line to "predict" the hidden 20%, which already happened.
Step 4. Compare the predictions with what really happened → R^2, RMSE, MAE.
Step 5. Rebuild the line using 100% of the data for the real forecast.

This mimics real use: predicting weeks the model has never seen. If there are fewer than 10 data points, there is too little to hide, so AutoVet checks against all the data and labels it *"in-sample"*.

3. The three measures in plain words

Measure	Question it answers	Simple meaning	Good when
R^2	Does the line follow the pattern?	A score from 0 to 1. 0.85 means the line explains 85% of the ups and downs.	Close to 1
MAE	On average, how far off are we?	The average miss, in real units (e.g. "off by 1 appointment").	Close to 0
RMSE	Are there any big misses?	Like MAE, but big misses count extra. If RMSE is much larger than MAE, a few predictions were far off.	Close to 0

The formulas

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2} \qquad \text{MAE} = \frac{1}{n}\sum|y_i - \hat{y}_i| \qquad \text{RMSE} = \sqrt{\frac{1}{n}\sum(y_i - \hat{y}_i)^2}$$

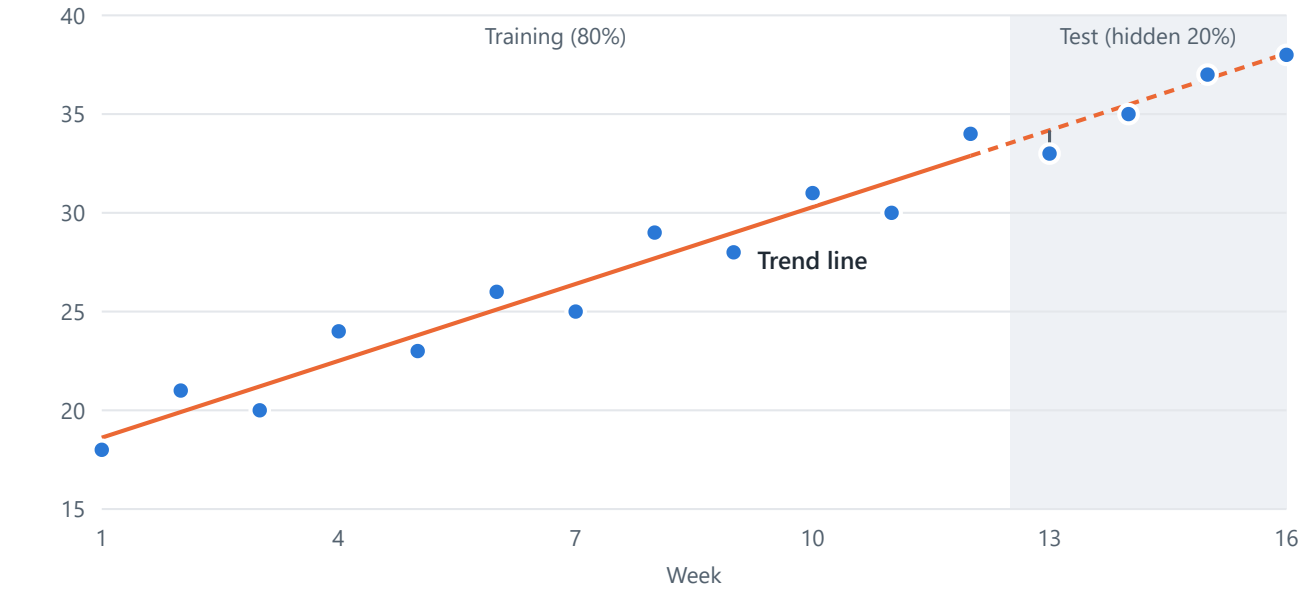
y_i = actual value, $\hat{y}_i$ = predicted value, $\bar{y}$ = average of the actual values, n = number of values checked. The difference $y_i - \hat{y}_i$ is the **error** (the miss).

4. Worked example: weekly appointments

> **Sample data.** These numbers are made up for illustration and are not real clinic records. The method is exactly what AutoVet does: 16 weeks of history, split into 12 training weeks and 4 test weeks.

The data

Week	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Appointments	18	21	20	24	23	26	25	29	28	31	30	34	33	35	37	38



● Actual appointments — Trend line learned from weeks 1–12 (dashed = prediction for hidden weeks)

Step 1. Learn the trend from weeks 1–12

$$\hat{y} = 1.30\,x + 18.62$$

In plain words: appointments grow by about **1.3 per week**. (*x* counts weeks from 0.)

Step 2. Predict the hidden weeks 13–16 and measure each miss

Week	Actual y	Predicted $\hat{y}$	Error $y - \hat{y}$	Error²	|Error|
Week 13	33	34.18	−1.18	1.40	1.18
Week 14	35	35.48	−0.48	0.23	0.48
Week 15	37	36.78	+0.22	0.05	0.22
Week 16	38	38.07	−0.07	0.01	0.07
Total				**1.68**	**1.96**

Values are rounded to 2 decimals for display. Totals use unrounded values.

Step 3. Compute the three measures

MAE: average of the |errors|

$$\text{MAE} = \frac{1.18 + 0.48 + 0.22 + 0.07}{4} = 0.49$$

RMSE: square the errors, average them, take the square root

$$\text{RMSE} = \sqrt{\frac{1.40 + 0.23 + 0.05 + 0.01}{4}} = \sqrt{0.42} = 0.65$$

R^2: compare our errors with the spread of the real values (average of weeks 13–16 = 35.75)

$$\sum (y_i - \bar{y})^2 = (33 - 35.75)^2 + (35 - 35.75)^2 + (37 - 35.75)^2 + (38 - 35.75)^2$$

$$= 7.5625 + 0.5625 + 1.5625 + 5.0625 = 14.75$$

$$R^2 = 1 - \frac{1.68}{14.75} = 0.89$$

Result

R^2

0.89

The line explains about 89% of the week-to-week changes. On AutoVet's confidence scale this is **High** (≥ 0.70).

MAE

0.49

On average the forecast missed by about **half an appointment** per week.

RMSE

0.65

Close to MAE, so there were **no big surprise misses**.

Is that actually good? Compare with simply guessing the average

Suppose instead we predicted 35.75 appointments (the average) every week:

Method	R^2	MAE	RMSE
Guess the average every week	0.00	1.75	1.92
AutoVet trend line	**0.89**	**0.49**	**0.65**

The trend line's average miss is about **3.6× smaller** than the guess. This is exactly what R^2 measures: R^2 = 0 means "no better than guessing the average", and R^2 = 1 means a perfect prediction.

5. How to read AutoVet's numbers

R^2	AutoVet confidence label	What to tell the clinic
0.70 – 1.00	**High**	Usage is steady; the forecast can be relied on for planning.
0.40 – 0.69	**Medium**	Usable, but double-check before big orders.
below 0.40	**Low**	Usage is irregular; collect more data before trusting the forecast.

MAE and RMSE have no fixed "good" cut-off. They are in real units, so judge them against the size of the numbers: a miss of 1 appointment is small when the clinic sees 35 a week, but big if it sees 3.

Where these appear in the system

Forecast	What is predicted	Measures reported
Appointment forecast (Dashboard)	Appointments per week, from the last 16 weeks	R^2, test R^2, RMSE, MAE
Inventory forecast	Total quantity used over time, per item, giving days until stock-out	R^2 (confidence), test R^2, RMSE, MAE

6. Likely panel questions

Why linear regression?

Clinic data (weekly appointments, daily usage of supplies) mostly follows a steady trend. A straight line is simple, fast, needs little data and is easy to explain: the slope literally means “appointments per week” or “units used per day”.

Why split 80/20 by time instead of randomly?

Forecasting is about the future. Testing on the most recent weeks copies real use: learn from the past, predict what comes next. A random split would let the model “peek” at future weeks.

Why report three measures instead of one?

Each answers a different question. R^2 shows if the pattern is captured, MAE shows the typical miss in real units, and RMSE warns about large misses. Together they give a complete picture.

What if R^2 is low?

It means usage is irregular (for example, a seasonal item or a new product). AutoVet marks the forecast as **Low** confidence and advises collecting more data, so staff don't rely on it blindly.

What if there is very little data?

With fewer than 10 data points there is not enough to hide 20%, so the measures are computed on all data (“in-sample”), and the system shows this in its validation method.

After testing, why retrain on 100% of the data?

The 80/20 split is only for checking accuracy. For the real forecast we want the model to learn from every week available, especially the most recent ones.